



Introduction to jQuery

1. Need of jQuery
2. Advantages of jQuery
3. jQuery selectors
4. jQuery events
5. Adding jQuery functionality
6. document.ready event

- Introduction and features
 - jQuery is javascript library
 - Simplifies JavaScript programming
 - Provides lot of functions needed for all the tasks to be carried on HTML page
- Tasks which are simplified
 - Traversing the the DOM tree
 - Modifying HTML content
 - Change styles of HTML element
 - Applying effects on HTML elements
 - AJAX calls

- History
 - emerged in 2006 developed by John Resig
 - Initially focused on DOM manipulation and event handling
 - from 2007 jQuery widely used by web developers

- Easy to learn
 - Anybody with the basic working knowledge of HTML, CSS and JavaScript can start development with jQuery.
- Save lots of time
 - inbuilt functions for traversing HTML, modifying HTML and applying different effects on HTML elements saves development efforts
- Simplify common JavaScript tasks
 - It will be very easy to create feature rich and interactive web pages with fewer lines of codes, a typical example is implementing Ajax to update the content

- Cross-browser support
 - It is compatible with all major modern browsers such as Chrome, Firefox, Safari, Internet Explorer, etc.
- Absolutely Free
 - Free to download and use.
- Performance
 - jQuery is lightweight and efficient which minimizes page load times.

- Selecting the elements using jQuery selectors is very easy.
- jQuery supports almost all the selectors defined in the latest CSS3 specifications
- Following are some examples of jQuery selectors
 - Selecting all: `$("*")`
 - Selecting current: `$(this)`
 - Selecting using id: `$("#userid")`
 - Selecting using classes: `$(".content")`
 - Selecting using tag names: `$("p")`
 - Selecting using attributes: `$("[type='text']")`

- Group selection: `$("p, a, span")`
- Child selection: `$("div > p")`
- Descendant selection: `$("div p")`
- Sibling selection: `$("p + a")`
- Pseudo class selection: `$(":focus")`

- It provides a comprehensive set of event handling methods to interact with user actions and DOM changes.
- Event methods need a callback function as the argument which defines the task to be performed on occurrence of event
- Different categories of events are
 - Document events:
 - i. `load()`
 - ii. `ready()`
 - Mouse events:
 - i. `mouseup()`
 - ii. `mousedown()`
 - iii. `click()`
 - iv. `mouseenter()`
 - v. `mouseleave()`

- Keyboard events
 - i. `keyup()`
 - ii. `keydown()`
 - iii. `keypress()`
- Form events
 - i. `blur()`
 - ii. `focus()`
 - iii. `change()`
 - iv. `submit()`

- jQuery library needs to be made available to the HTML page. There are two methods to include jQuery in the existing HTML page
- Download and include jQuery file
 - You can download jQuery library on your local machine. There are two versions of jQuery available for downloading:
 - i. Production version - this is for your live website because it has been minified and compressed
 - ii. Development version - this is for testing and development (uncompressed and readable code)
 - specify the jquery path in HTML file using script tag.

- Including the jQuery by CDN
 - Use CDN path for src attribute of `<script>` tag

Note : Advantage of using the hosted jQuery from Google as many users already might have downloaded jQuery when visiting another site. This leads to faster loading time.

- This is to prevent any jQuery code from running before the document is finished loading (is ready).
- It is good practice to wait for the document to be fully loaded and ready before working with it.
- All jQuery methods are inside a document ready event like

```
$(document).ready(function(){  
    // jQuery methods go here...  
});
```